

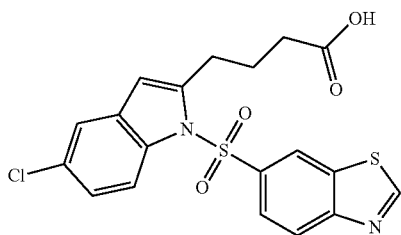
SOLID STATE FORMS OF LANIFIBRANOR AND PROCESS FOR PREPARATION THEREOF

FIELD OF THE DISCLOSURE

[0001] The present disclosure encompasses solid state forms of Lanifibranor, in embodiments crystalline polymorphs of Lanifibranor, processes for preparation thereof, and pharmaceutical compositions thereof.

BACKGROUND OF THE DISCLOSURE

[0002] Lanifibranor, 1-(6-benzothiazolylsulfonyl)-5-chloro-1H-indole-2-butanolic acid, has the following chemical structure:



[0003] Lanifibranor is a pan peroxisome proliferator-activated receptor, and it is developed for the treatment of non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease, type 2 diabetes, fibrosis and systemic sclerosis.

[0004] The compound is described in International Publication No. WO2007/026097.

[0005] Polymorphism, the occurrence of different crystalline forms, is a property of some molecules and molecular complexes. A single molecule may give rise to a variety of polymorphs having distinct crystal structures and physical properties like melting point, thermal behaviors (e.g., measured by thermogravimetric analysis ("TGA"), or differential scanning calorimetry ("DSC")), X-ray diffraction (XRD) pattern, infrared absorption fingerprint, and solid state (^{13}C) NMR spectrum. One or more of these techniques may be used to distinguish different polymorphic forms of a compound.

[0006] Different salts and solid state forms (including solvated forms) of an active pharmaceutical ingredient may possess different properties. Such variations in the properties of different salts and solid state forms and solvates may provide a basis for improving formulation, for example, by facilitating better processing or handling characteristics, changing the dissolution profile in a favorable direction, or improving stability (polymorph as well as chemical stability) and shelf-life. These variations in the properties of different salts and solid state forms may also offer improvements to the final dosage form, for instance, if they serve to improve bioavailability. Different salts and solid state forms and solvates of an active pharmaceutical ingredient may also give rise to a variety of polymorphs or crystalline forms, which may in turn provide additional opportunities to assess variations in the properties and characteristics of a solid active pharmaceutical ingredient.

[0007] Discovering new solid state forms and solvates of a pharmaceutical product may yield materials having desirable processing properties, such as ease of handling, ease of

processing, storage stability, and ease of purification or as desirable intermediate crystal forms that facilitate conversion to other polymorphic forms. New solid state forms of a pharmaceutically useful compound can also provide an opportunity to improve the performance characteristics of a pharmaceutical product. It enlarges the repertoire of materials that a formulation scientist has available for formulation optimization, for example by providing a product with different properties, including a different crystal habit, higher crystallinity, or polymorphic stability, which may offer better processing or handling characteristics, improved dissolution profile, or improved shelf-life (chemical/physical stability). For at least these reasons, there is a need for additional solid state forms (including solvated forms) of Lanifibranor.

SUMMARY OF THE DISCLOSURE

[0008] The present disclosure provides crystalline polymorphs of Lanifibranor, processes for preparation thereof, and pharmaceutical compositions thereof. These crystalline polymorphs can be used to prepare other solid state forms of Lanifibranor, Lanifibranor salts and their solid state forms.

[0009] In embodiments, the present disclosure provides Lanifibranor and crystalline forms thereof. In embodiments, the present disclosure provides crystalline forms of Lanifibranor designated as Form LN1 and Form LN2 (defined herein).

[0010] The present disclosure also provides uses of the said solid state forms of Lanifibranor in the preparation of other solid state forms of Lanifibranor or salts thereof.

[0011] The present disclosure provides crystalline polymorphs of Lanifibranor for use in medicine, including for the treatment of non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease, type 2 diabetes, fibrosis and systemic sclerosis.

[0012] The present disclosure also encompasses the use of crystalline polymorphs of Lanifibranor of the present disclosure for the preparation of pharmaceutical compositions and/or formulations.

[0013] In another aspect, the present disclosure provides pharmaceutical compositions comprising crystalline polymorphs of Lanifibranor according to the present disclosure.

[0014] The present disclosure includes processes for preparing the above mentioned pharmaceutical compositions. The processes includes combining any one or a combination of the crystalline polymorphs of Lanifibranor with at least one pharmaceutically acceptable excipient.

[0015] The crystalline polymorph of Lanifibranor as defined herein and the pharmaceutical compositions or formulations of the crystalline polymorph of Lanifibranor may be used as medicaments, such as for the treatment of non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease, type 2 diabetes, fibrosis and systemic sclerosis.

[0016] The present disclosure also provides methods of treating non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease, type 2 diabetes, fibrosis and systemic sclerosis, by administering a therapeutically effective amount of any one or a combination of the crystalline polymorphs of Lanifibranor of the present disclosure, or at least one of the above pharmaceutical compositions, to a subject suffering from non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease, type 2 diabetes, fibrosis and systemic sclerosis, or otherwise in need of the treatment.

Figure 4: X-Ray Powder Diffraction Pattern of Lanifibranor Form LN3

